

Grattan's Consumer Energy Wallet Analysis – Methodology Paper

1. Overview

The Grattan's forthcoming report will make the case for a carbon price in the electricity market. To do this, we need to understand the effect on consumers. The report will rely on external modelling to produce wholesale and retail electricity prices under a range of policy scenarios. To understand how this translates to consumer bills requires further analysis.

Our approach looks at the household “energy wallet” (Gas, petrol, and electricity) rather than just electricity costs, to capture the changing way households consume energy over the transition. This analysis uses consumer usage data and AEMO forecasts to estimate household electricity, gas, and petrol consumption over time. Consumption data is then combined with price forecasts to understand consumer energy costs over time.

The energy wallet analysis will make clear the cost impacts for consumers across their energy consumption and provide estimates for the savings associated with electrification and access to consumer energy resources (Rooftop PV, batteries).

The energy wallet analysis represents annual usage costs and does not include capital costs for appliances. Separate pay off period analysis for rooftop PV and batteries will be included.

The report will use two approaches to estimate household energy costs

I. Average household / Total household costs

- This consumer represents the sum of gas, electricity, and petrol consumption by jurisdiction and year, divided by the number of electricity connections.
- This consumer will represent the energy costs faced by the community as a whole, rather than attempting to model a “real” household.

II. Household cameos

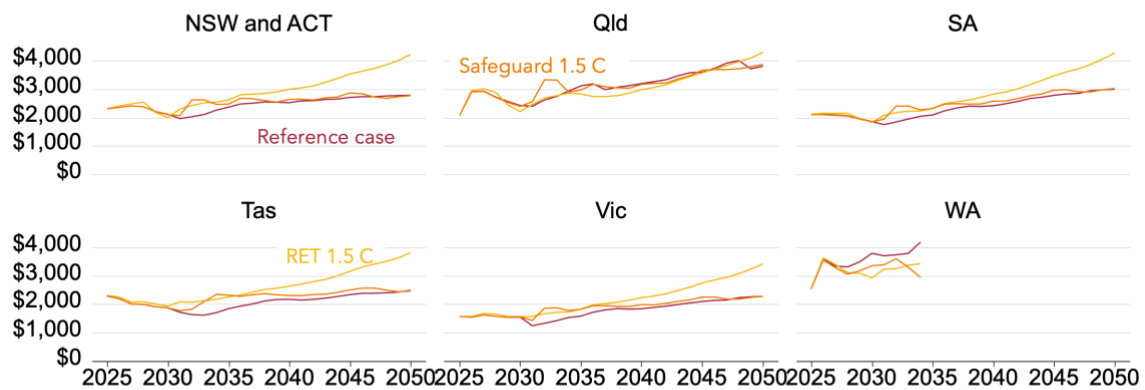
- These consumers will represent multiple household types broken down by EV ownership, degree of electrification / use of gas, and access to PV
- For example,
 - Household (a) is fully electric, owns 1 EV, has rooftop PV, and a household battery.
 - Household (b) uses gas cooking and heating, owns 1 ICE car, has rooftop PV, and no battery.

Some example charts to give you a feel for the outputs of this project:

Household electricity bills are set to rise across scenarios

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Household electricity bills (\$2025)



... But total household energy costs are set to decline

Total household energy bills (\$2025)

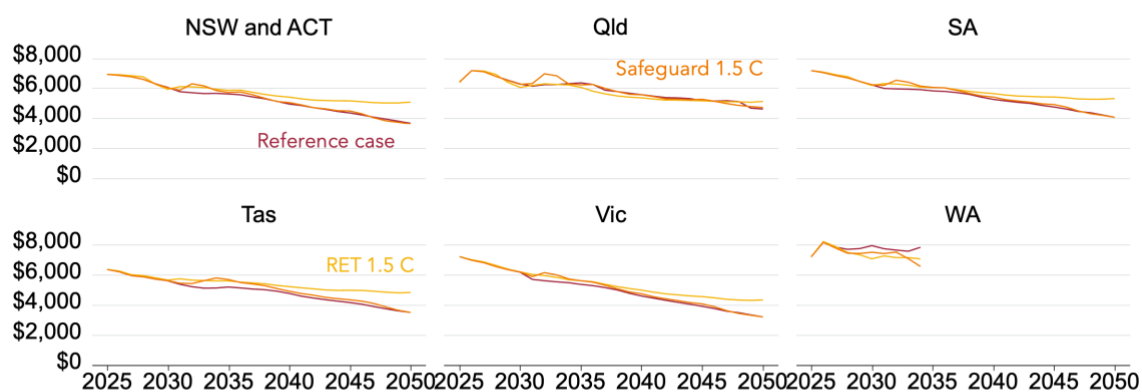
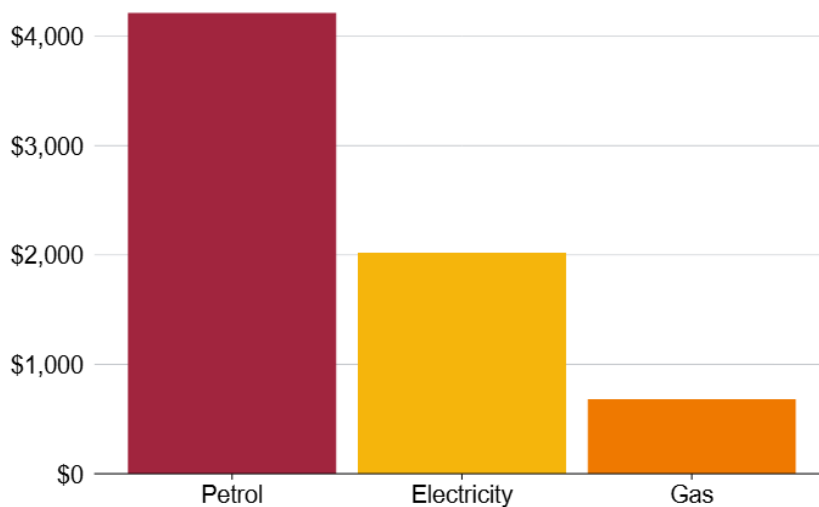


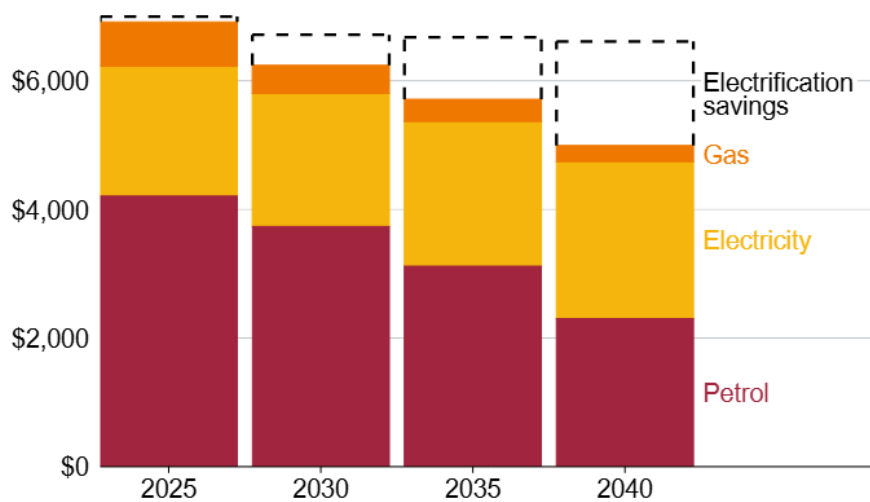
Figure 5.1: Electricity forms just part of total household energy costs
Average household energy costs by fuel source



Notes: Grattan analysis. See ?? for further details.

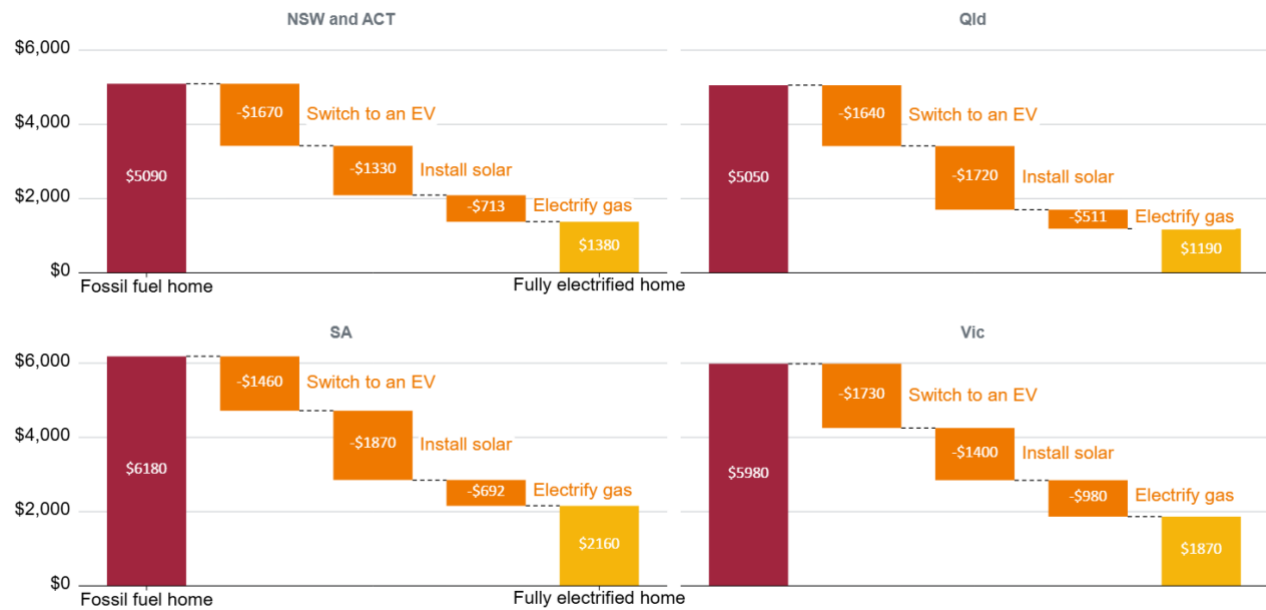
Figure 5.2: On average, electrification will save Australian households \$1,620 a year by 2040

Average annual household energy costs, by fuel source (\$2024)



Notes: Grattan analysis. See ?? for further details.

Figure 5.3: Households can access large savings with electrification and solar
Annual household energy costs, by consumer type (\$2024)



Notes: Grattan analysis. See ?? for further details.

2. Method

Projecting energy prices

All prices are expressed in 2025 dollars.

Electricity

The start point for retail electricity tariffs is the best offer from a big 3 retailer in each jurisdiction. Retail tariffs were sourced from:

- NEM <https://www.energymadeeasy.gov.au/>
- Vic <https://compare.energy.vic.gov.au/>
- WA <https://wattever.com.au/compare-best-electricity-rates/#nsw>
- Tas <https://wattever.com.au/compare-best-electricity-rates/#nsw>

In July of 2025.

These tariffs were then scaled according to the retail price projections by Jacobs. These prices were broken down by region/jurisdiction and year and represent a flat rate per MWh.

Feed-in tariffs by jurisdiction are provided by Jacobs and reflect current retail feed-in tariffs scaled down to zero by 2030. This reflects expectations that the wholesale value of exports will decline to zero by that time and no further government subsidies will be applied.

Relevant scripts	Relevant data sources
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• Get_jacobs_retail_prices.R • get_electricity_tariffs.R	• Jacobs modelled retail prices ◦ RetailPriceProjections files for all scenarios stored in Data/Jacobs • Data/electricity_tariffs_july_25.xlsx
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Gas

Volumetric charges:

To project volumetric gas prices for residential customers, we start by taking the best offer from the big 3 retailers (AGL, Energy Australia, and Origin Energy) in each jurisdiction in July of 2025. The ‘best’ offer was determined by the reference price used by Energy Made Easy and Compare Energy.

Relevant scripts	Relevant data sources
• get_gas_best_offers.R	• Data/gas_offers.xlsx, sheet we_best_tariffs ◦ https://wattever.com.au/compare-best-gas-rates/ ◦ https://www.energymadeeasy.gov.au/ ◦ https://compare.energy.vic.gov.au/

The volumetric component of the gas bill was then projected out to 2050 by indexing it to the forecast natural gas price. The natural gas price forecast for residential and commercial customers is taken from estimates used by [ACIL Allen for the 2024 Gas Statement of Opportunities \(GSOO\)](#) and the WA GSOO and expressed in real 2025 Australian dollars. This represents a simplistic assumption that volumetric costs rise proportionally to the industrial gas price, and that changes in usage of the gas network are solely passed through in network charges.

Relevant scripts	Relevant data sources
• get_gas_prices_data.R • project_gas_retail_volumetric_price.R	• ACIL Allen for the 2024 Gas Statement of Opportunities (GSOO) • 2024 Natural Gas Price Forecasts workbook - Western Australia - Expected case - Industrial

Network charges:

To project gas network charges, we assume that network charges increase inverse proportionally to the decline in gas customers (i.e. the total revenue recovered through network charges is held constant in real terms) in each state over the modelled period. This assumption is approximate only, and doesn’t take into account changes in the network asset base or in maintenance expenditure as overall gas consumption declines.

The total network charges revenue for each state are estimated as the annual network charge for the best offer in each state, multiplied by the total residential customers in each state. Total

residential customers in each state are sourced from AER retail performance reporting data (for NEM states other than Vic) and the ESC annual overview data (for Vic).

Relevant scripts	Relevant data sources
• <code>get_gas_connections_data.R</code> • <code>calculate_total_gas_network_revenue.R</code>	• https://www.aer.gov.au/documents/schedule-2-quarter-1-2024-25-retail-performance-data • https://www.aer.gov.au/documents/schedule-2-quarter-3-2023-24-retail-performance-data • https://www.esc.vic.gov.au/electricity-and-gas/victorian-energy-market-reporting-hub/victorian-energy-market-insights/energy-market-dashboard (FY24) • Tasmanian Economic Regulator • Economic Regulation Authority WA

To project out network charges per residential gas connection, we first project residential gas connections using the 2024 gas statement of opportunities (GSOO). The 2024 GSOO provides **total small customer consumption** out to 2043. Consumption to 2050 is a linear projection of the 2030 to 2043 trend in each state. For Victorian consumption, a linear projection leads to negative gas consumption in 2050. To correct for this, we assume that Victorian gas consumption is constant from 2049 to 2050.

The proportion of 2023 **total small customer consumption** that is residential is estimated by multiplying the **benchmark consumption per connection** used above (2020 Frontier Economics) by the number of residential connections in 2023. This provides an estimate of **total residential consumption** in each state and is divided by the **total small customer consumption** to derive the proportion of small customer consumption that is attributable to residential connections. This proportion is assumed to be constant over the modelled period.

The projected number of connections is then derived by assuming the gas consumption per connection is constant. That is, **projected gas connections** = **total residential consumption** / **benchmark consumption per connection**. The **benchmark consumption per connection** is sourced from the 2020 Frontier Economics benchmark consumption data. In reality, average gas consumption per connection is declining slightly as partial electrification occurs.

Finally, the projected network charge per residential gas connection is calculated by dividing the **total gas network revenue** by the **projected number of residential gas connections** for each year.

Relevant scripts	Relevant data sources
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• project_residential_gas_consumption.R • project_gas_connection_charges.R	• residential energy consumption benchmarks prepared for the AER in 2020 by Frontier Economics. • https://aemo.com.au/energy-systems/gas/gas-forecasting-and-planning/gas-forecasting-data-portal (2024) • WA GSOO 2024
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Petrol

Petrol prices are sourced from the [ACCC 2025 March quarterly report on the petroleum market](#) and are held constant in real terms throughout the projection.

Relevant scripts	Relevant data sources
• get_petrol_data.R	• ACCC 2025 March quarterly report on the petroleum market figure 1.11

Household connections, Internal combustion engine (ICE) and EV data – used across modules

Household connections

Household electricity connections projections are taken from AEMO (ESOO 2024 and WEM ESOO 2024). To estimate the number of connections between 2020 and 2025 (when AEMO projections start) we use ABS dwelling completions data.

To get SWIS (WA) connections prior to 2024 we use Synergy performance reporting documents.

Relevant scripts	Relevant data sources
• get_household_connections_data.R	• ESOO 2024 Assumptions Workbook • WEM ESOO 2024 Data Register • ABS Dwelling completions • Synergy performance reporting ○ https://www.synergy.net.au/-/media/Files/PDF-Library/Synergy-electricity-retail-licence-ERL1-performance-report-datasheet-2022_2023.pdf ○ https://www.synergy.net.au/-/media/Documents/Policies/2022-Synergy-Electricity-Retail-Licence-Performance-Reporting-Datasheet.pdf

	○ https://www.synergy.net.au/-/media/Documents/Policies/2021-Electricity-Retail-Licence-Performance-Reporting-Datasheet---Synergy.pdf ○ https://www.synergy.net.au/-/media/Documents/Policies/2020-Electricity-performance-reporting.pdf
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EV data

We take EV fleet projections from the 2024 (NEM) ESOO and 2024 WEM ESOO to determine the proportion of the total residential vehicle fleet that is an EV over time.

We also take the time of use electricity consumption of EV vehicles from these same projections to calculate the transport electricity load for households with an EV.

Demand traces for EVs in the ESOO are based on vehicle use and charging patterns in 2040. For the WEM, these are based on 2030 use and charging. As EV efficiency is not forecast to change significantly over time, these profiles are suitable proxies for all years.

For the purpose of consumer bills, all EV charging is assumed to take place at home so that the costs are included in the household bill. Actual costs may be slightly higher if vehicles are charged outside of the home.

Relevant scripts	Relevant data sources
• Get_aemo_vehicle_data.R • Get_ev_consumption_profiles.R	• 2024 ESOO electric vehicle workbook • 2024 WEM ESOO electric vehicle projections

Petrol consumption

The average number of motor vehicles per household is 1.8

<https://www.abs.gov.au/statistics/industry/tourism-and-transport/transport-census/2021>

Vehicles are assumed to be driven the average number of kilometres as described by the [2020 survey of motor vehicle use](#). Fuel efficiency is also described by the survey (0.111 Litres/km). Fuel consumption per vehicle has remained flat over the past decade as any efficiency gains have been compensated for by larger vehicles. We assume that the rate is constant throughout the period, but we may want to defensively add a decline rate e.g 1% per annum.

Relevant scripts	Relevant data sources
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• <code>get_average_petrol_use_per_km.R</code> • <code>average_km_per_vehicle.R</code>	• 2020 survey of motor vehicle use.
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Household cameo consumption

Household cameos were constructed by the variables listed below. The model produces outputs for a household with any combination of values. Savings can be calculated for each change in variable.

Variable	Possible values	Example Household A	Example Household B
Cooking	{Gas, Electric}	Electric	Gas
Space conditioning	{Gas, Electric}	Electric	Electric
Water heating	{Gas, Electric}	Electric	Gas
PV	{True, False}	True	True
Battery	{True, False}	True	False
Electric vehicles	All positive real numbers.	1	0
ICE vehicles	All positive real numbers	0	2
State	All states and territories, NSW and ACT are aggregated.	NSW and ACT	Qld

Household baseline fuel consumption

The baseline for total household gas and electricity consumption is taken from the Residential Baseline Study (2022). Data for 2020 in the study represents a real, bottom up estimate of total fuel consumption in each jurisdiction for the existing appliance and housing stock. Estimates of electricity and gas consumption per household are derived by dividing the total consumption by the number of households.

The number of households is deflated by 10% to represent the assumed occupancy rate of the existing stock. This assumption is used by the RBS to derive total usage for each jurisdiction.

To estimate the fuel usage of households with different appliance mixes, we apply a conversion coefficient to the usage of one fuel type and convert that usage to the other fuel type.

E.g. electric heaters are X% more efficient than gas heaters. Heating usage if all heating was served by electric heaters = current heating electricity usage + current gas usage * 1/X.

Time of use electricity consumption data is also taken from the RBS. These traces are then scaled according to calculated electricity consumption for each end use for each consumer type.

As almost zero cooling is served by gas appliances, we need to separate the aggregate heating and cooling load. This is done using typical temperature data for each state.

Electric appliance efficiency over time is derived from AEMO residential energy efficiency forecasts for the NEM and WA. Gas appliances are assumed to not improve in efficiency, consistent with other approaches including AEMC price trends.

Set up scripts:

load_and_deflate_rbs_households.R

- Loads household data and applies the 90% occupancy factor.

get_rbs_fuel_end_use.R

- Loads RBS total fuel consumption by end use (heating, cooking, etc.) and jurisdiction

Get_fuel_conversion_coefficients.R

- This script loads in our conversion coefficients between gas and electricity

The logic behind these coefficients has already been QC'd by Ben Jefferson and represent the most common appliance types for each end use.

End use	Gas to electric coefficient	Interpretation	Source(s)
Cooking	0.43	The mean electric oven and cooktop consumption relative to a gas oven and cooktop	Frontier Energy 2019
Water heating	0.81	Represents electric storage energy consumption relative to gas instantaneous	Climate Council 2022 Table 3
Space heating	0.19	Represents RCAC heating relative to ducted gas. RCAC CoP = 3.65 Ducted gas CoP = 0.88, ducted losses = 20%	Climate Council 2022 IEEFA 2025 , pg 5 Climate Choices

Hot water

- Gas instant and electric storage are the two most common appliance types for each fuel type (RBS 2022)
 - Electric storage is currently far, far more common than heat pumps (not suitable)

Space conditioning

- AC non-ducted and Gas ducted are the two most common appliance types for each fuel type (RBS 2022)

Cooking

- Appliance choice and efficiencies are straightforward,

get_rbs_electricity_consumption_data.R

- Loads in the RBS time of use electricity consumption data
- For space conditioning in Victoria, where the vast majority of heating is served by gas. For a more accurate profile, Mohseni et al. [Residential load profiles for Heyfield, Victoria](#). May 2023. Was used instead of the RBS profile.

load_temperature_data.R

- This script loads in typical meteorological year weather data from CSIRO 2024
- <https://ahd.csiro.au/wp-content/uploads/2021/08/Typical-meteorological-year-weather-files-User-Guide-v4.pdf>
- <https://agdatashop.csiro.au/future-climate-typical-meteorological-year>

calculate_space_heating_tou.R

- This script disaggregates heating and cooling loads by estimating the proportion of the load that is heating or cooling in each season in a typical year.
- Heating and cooling loads are typically proportional to the heating degree days (HDDs) and cooling degree days (CDDs) in that location respectively.
- HDDs and CDDs are calculated by multiplying the difference between the average temperature of that day and the comfort temperature (<http://www.bom.gov.au/climate/maps/averages/degree-days/>)
 - E.g. av temp of day = 15, comfort temp = 18, Heating degree days = 18-15 = 3
 - The actual proportion of the load will vary as behaviour is often different for heating and cooling and appliance efficiencies are also slightly different, but it is a good approximation to determine how much of our aggregated load in each season is used for heating and how much is used for cooling.

Calculate_fuel_use_conversions.R

- This script takes the total fuel use for each end use and calculates the total consumption for each category if the fuel was switched from gas to electricity or vice versa

Calculate_household_energy_efficiency.R

- This script estimates the efficiency gains for electrical appliances from 2020 to 2050 using forecasts from AEMO for the NEM and the WEM
- AEMO forecasts include total demand reductions from energy efficiency as well as total underlying residential demand. We can estimate the reduction in residential demand

from efficiency gains by dividing the underlying demand with efficiency gains by the underlying demand without efficiency gains (underlying demand / (underlying demand + energy efficiency))

- Doing so gives us an index that we can use to scale down residential electricity consumption in our base year of 2020 down to the demand in 2050.

Creating profiles

We can now create total consumption and ToU demand profiles for each of our consumer types using the converted fuel consumption estimates, PV data, battery data, and EV data.

rbs_fuel_consumption_profiles.R

- This script calculates the estimated fuel consumption for each end use if the total RBS fuel use in that state was converted from a mix of fuels to one fuel type (gas or electricity)
- we can then combine these and divide by the number of households to create profiles for each consumer type (e.g gas cooking, electric heating, electric water heating)

get_pv_profiles.R

- This script reads in PV data generated for each capital city using the [PV Watts calculator](#).
- Data is generated for both 7kw and 10 kw systems. For the cameo profiles, it is assumed that a typical 7 kw system is installed following the [AEMC Price Trends](#) assumption.
- Capacity factors (actual generation/total possible generation) are then applied for each state using estimates from [CSIRO 2024](#) table A-1

Calculate_tou_consumer_profiles.R

- This script combines all of our data to generate time of use electricity profiles for each consumer type.
- First, we normalise the bas tou profiles from the RBS such that the profiles, when summed over a year, equal 1.
- Then, we scale the profiles by actual annual energy consumption as calculated in rbs_fuel_consumption_profiles.R
- Next, we apply efficiency gains to extrapolate our 2020 consumption out to 2050
- Finally, we add in the EV tou profiles and PV profiles calculated earlier

Generate_battery_profiles.R

- This script runs a simple algorithm on our time of use profiles to simulate the behaviour of a battery that simply charges when there is excess solar and discharges when the battery has charge and the household would otherwise need to import electricity.
 - This will underestimate the value of battery installation as it doesn't allow the consumer to take advantage of time of use tariff arbitrage or participate in FCAS markets. CSIRO estimates that

- The battery profile shape is not suitable to be assessed against ToU tariffs for this reason, but is reasonably accurate to estimate the amount of electricity imports that can be avoided with a battery.
- Assumptions taken from [CSIRO 2024](#) table A-2:
 - Round trip efficiency = 85%
 - Maximum charge of discharge of rated capacity = 85%
 - Capacity = 11 kWh
 - Maximum power (maximum amount that can be discharged in one hour for our purposes) = Rated capacity divided by 2.1
- Battery capacities are not degraded over time (nor is solar capacity) so profiles are only suitable for point in time comparisons rather than over the lifetime of the appliance.
 - This can easily be corrected for using 1.8% degradation per annum on kWh capacity from CSIRO 2024 for lifetime comparisons.
 - For e.g 10 year costs charts and payback periods we will add this degradation in
- For internal consistency, we ensure that the battery finishes each day with same amount of charge that it started with. As we are only running the algorithm over a single day, this makes sure that we are not inflating the discharge capacity of the battery at the start of the simulation.
 - To do this, we create a function that runs our battery according to the simple charge-discharge algorithm
 - We then run optimise() to find the starting charge level that minimises the difference between the start charge and the end charge
- Once we have determined the starting charge level, we run the algorithm from that start charge and calculate the behaviour of the battery for each household profile.

calculate_annual_electricity_consumption_averages.R

- This script takes our time of use profiles and some the total consumption/generation in each hour to determine if a household is importing or exporting electricity in that hour (and how much).
- The total imports and exports are then summed over the year to produce annual values.

Calculating profile costs

Calculate_cameo_gas_costs.R

- This function takes total gas consumption for each consumer profile, applies an annual connection charge from our projections and applies the average volumetric charge from our projections

Calculate_cameo_electricity_costs.R

- This script takes the electricity consumption for each profile and applies our projected tariffs. These tariffs are all flat rate and are likely to underestimate the available benefits to consumers who can shift their time of consumption

Calculate_cameo_petrol_costs.R

- This function calculates the petrol costs to consumers that own 0,1, or 2 ICE vehicles

Average household consumption

The consumer profiles calculated above provide estimates of the different energy costs faced by different types of consumers, but they do not tell us anything about the average consumer. The average consumer represents the energy consumption and costs faced by the community as a whole. This module uses AEMO projections of residential gas and electricity usage to determine average household consumption over time.

While consumers are not net exporters on average, some consumers do export electricity, changing the costs they face. To correct for the effect that some households access solar and batteries to export to the grid, we segment households into consumer types: No solar, Solar, Solar and battery.

This module consists of 7 core steps:

- Calculate average underlying residential demand (this is household demand before any PV generation is netted out) over time from AEMO forecasts (AEMO separates this into baseline demand, EV demand, demand from electrification of gas, and averted demand from energy efficiency gains).
- Calculate the time of use load shape of baseline demand and electrification demand using the RBS (2022), and AEMO EV projections
- Generate average load profiles for consumers over time by scaling each ToU trace by each forecast component of demand (baseline and electrification)
 - Normalised ToU trace * forecast demand
- Allocate forecast rooftop solar generation to the forecast proportion of connections with solar.
- Allocate batteries to the forecast proportion of connections and reduce their consumption further with the simple battery algorithm
- Calculate bills for each consumer type and calculate average costs using a weighted average.
- Add in gas and petrol costs from AEMO forecasts

Average consumption calculations:

`get_esoo_average_underlying_demand.R`

- This script reads in the AEMO total underlying demand forecasts and divides by household connections to get average household demand.

`Calculate_average_load_shapes.R`

- This script calculates the electricity load shapes of the average household over time by splitting forecast demand into baseline demand (which follows the shape of the average household in 2020) and electrification demand (which follows the shape of cooking, water heating and space heating demand, weighted by the proportion of the total electrified load each end_use represents).

- E.g. in qld the electrified load is mostly gas cooking so the electrification profile has a similar shape to the QLD cooking profile
 - This assumes that the average household load shape in each state hasn't changed substantially between 2020 and 2024 which seems totally reasonable.
- First, we normalise the tou consumptions curves the same way we did for our cameos
- Then, we calculate weights for each end_use profile for each state by calculating the total electricity load that would exist if each end_use was completely converted from gas to electricity
 - Here we essentially assume that each end_use is electrified at the same rate, so the additional electrification load looks like the load if each end_use was declined by proportionally the same amount each year.
- Finally, we calculate an electrification load shape as the weighted average of the cooking, space heating and water heating load shapes

Estimate_pv_system_stock.R

- This script uses the 2020 RBS values for total number of PV systems and [CSIRO \(2024\)](#) estimates of PV prevalence by 2050 (**For Ben to look at**) to estimate the proportion of households with PV over time.
 - PV share estimates are based on CSIRO 2024 as only publicly available data, but generation is based on ESOO 2024 (CSIRO 2022). Estimates do not seem to be very different from public points of comparison (table 5-1 2022, table 5-1 2024). Should not substantially change results.
- Our projection is a simple linear interpolation between PV systems in 2020 to the 2050 estimates

estimate_battery_prevalence.R

- This script estimates the proportion of consumers that have access to the benefits of a battery using the forecast small scale embedded storage capacity projections from the 2024 ESOO.
- To estimate the number of consumers that contribute to the total capacity, we assume every battery has an 11kw rated capacity, following the average installation assumption of [CSIRO \(2024\)](#)
- Hence, battery stock = total capacity / 11
- We then divide the battery stock by the number of household connections to estimate the proportion of households with a battery over time

Calculate_average_pv_profile.R

- This script uses the projected residential PV generation from AEMO to scale our existing 7kw PV traces that we used in the cameos module
- First, we calculated the annual pv generation per system using our stock estimate
- Then, we calculate an implicit system size / scaling factor by comparing to annual generation of our 7kw profiles in each state.
- We then apply the scaling factor to our traces to calculate the average solar trace for a solar owning household over the projection.

calculate_average_profiles.R

- This script joins our normalised ToU profiles with our demand projections to get ToU demand over time.
- EV loads and PV are also added in
 - To estimate the average EV demand per household, we use the ToU demand trace per vehicle from AEMO
 - Households are assumed to own 1.8 motor vehicles (the national average, see source in earlier section) and the proportion of their vehicles that are an EV is determined by the forecast proportion of the fleet that are EVs from AEMO
- We join the demand traces together to create 4 profiles:
 - Household without PV – electrification occurs (gas, EV demand changes)
 - Household with PV - electrification occurs
 - Household without PV – electrification does not occur (gas, EV demand stays equal to 2024)
 - Household with PV - electrification does not occur
- This way we can (a) weight average costs by the number of households with PV and (b) compare consumer costs to a counterfactual scenario where electrification of gas and petrol does not occur

average_profiles_w_batteries is then calculated using the generate_battery_profiles function discussed above. All_average_profiles is simply the baseline profiles plus the profiles with batteries.

calculate_annual_electricity_consumption_averages.R

- This script works the same as calculate_annual_electricity_consumption_profiles.R discussed above and should really be the same function, sorry.

calculate_average_residential_gas_consumption.R

- This script takes our projections for total residential gas consumption in each state and divides by the number of residential connections (electric, to represent total number of households).
- A separate timeseries is also created where average gas consumption per household is held constant. This timeseries represents costs to consumers if electrification of gas appliances does not occur.

calculate_average_petrol_consumption.R

- Average petrol consumption per household is calculated by assuming 1.8 motor vehicles per household and the number of ICE vehicles is equal to $1.8 * \text{proportion of fleet that are ICE (AEMO forecasts)}$
- Again, a parallel dataset is created to represent the counterfactual in which EV uptake does not increase after 2024.

Calculate_average_gas_costs.R

- This script calculates average gas costs by multiplying average consumption by our volumetric price projections and dividing the total network charges revenue between all households.

calculate_average_electricity_costs.R

- Functions the same as calculate cameo electricity costs (sorry)

calculate_average_petroL_costs.R

- Functions the same as calculate cameo gas costs (sorry)